

Product Requirement Document: AI Communication & Public Speaking Coach

1. Executive Summary

An AI-powered communication and public speaking assessment application designed to help users refine scripts, improve vocal modulation, correct pronunciation, and enhance body language. The system operates on an asynchronous post-recording batch-processing model (max 2 minutes per session) to optimize compute cost and accuracy while providing cross-session continuity analysis over time.

2. Core Functional Requirements

2.1 Script Analysis & Tone Refinement

- **Input:** User uploads text script and selects target context (Formal, Semi-Formal, Casual).
- **Processing:** LLM evaluates script structure, word selection, and narrative flow.
- **Output:** Contextual diffs, alternative vocabulary recommendations, tone alignment scores, and continuity feedback.

2.2 Video & Audio Session Assessment (Batch Processing)

- **Input:** Camera and microphone recording up to 2 minutes per session.
- **Mode of Operation:** Asynchronous post-recording batch analysis (no real-time sub-second inference).
- **Audio Metrics:** Phoneme-level pronunciation alignment, vocal modulation (pitch variance), speaking rate (words per minute), pause cadence, and filler word detection.
- **Vision Metrics:** Eye contact ratio, head pose orientation, posture stability, and hand gesture frequency.

2.3 "Speak a Minute" Impromptu Mode

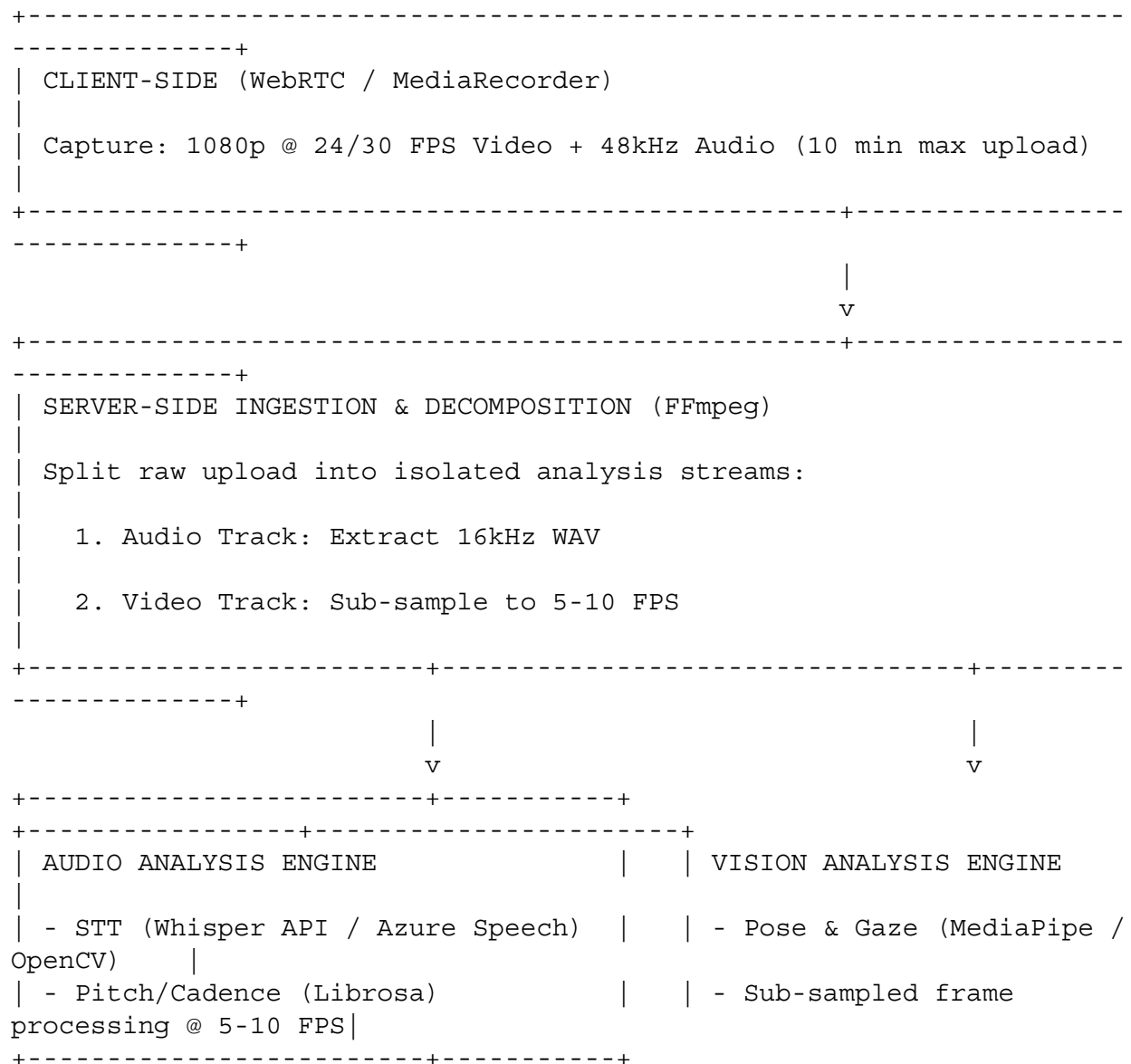
- **Topic Selection:** Topic roulette wheel spanning selected domains (Tech, Current Affairs, Business, General).
- **Preparation Window:** 10-minute timer for content structuring.
- **Research Assistant:** Instant key terms, research topics, and search queries generated by LLM pipeline to guide user preparation within the 10-minute window.
- **Evaluation:** Same video/audio assessment pipeline applied post-recording.

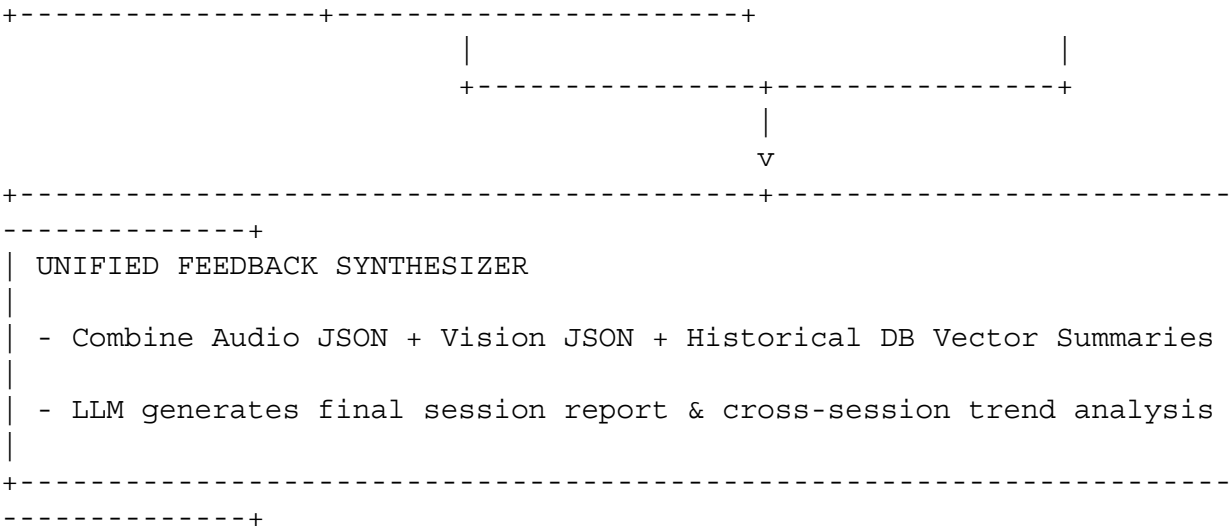
2.4 Pronunciation Repair Loop

- **Detection:** Forced-alignment STT isolates mispronounced words/phonemes.

- ## 2.5 Analytics & Cross-Session Continuity

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4. Technical Specifications & Performance Rules

Component	Specification	Rationale
Client Video Capture	24 FPS or 30 FPS @ 1080p	Smooth recording UX for user local preview.
Server Vision Pipeline	Adaptive Sub-sampling to 5-10 FPS	Optimizes processing cost; posture/eye contact change slowly over time.
Server Audio Pipeline	100% Full Resolution (16kHz WAV)	Phoneme isolation, pitch analysis, and STT require continuous waveform data.
Session Length Cap	10 Minutes Max	Caps cloud processing costs and server memory load.
Processing Model	Asynchronous Queue (Celery / Temporal + S3)	Avoids real-time WebRTC compute overhead while delivering deep multi-modal output.